Perturb-C and FFPE Workflows for Single-Cell 3D Genomics

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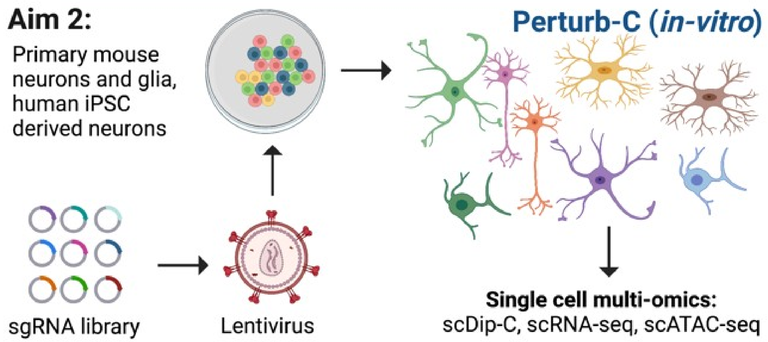
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Rationale and Research Goals

- Dip-C examines genome organization in individual cells. Perturb-C aims to test how genetic and chemical perturbations alter the 3D genome.
- The project focuses on aging-associated changes in the cerebellum, prefrontal cortex, and hippocampus.
- Key questions: Which proteins regulate genome organization? How do perturbations affect 3D structure and aging-associated phenotypes?

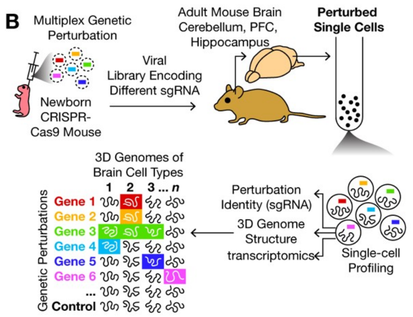
Perturb-C Screening Strategy

Proposed screens in cultured cells



Pooled CRISPR knockout screens in primary mouse cells and human induced pluripotent stem cell (hiPSC)-derived cells, with lentiviral guide delivery and single-cell readouts.

Proposed screens in mice



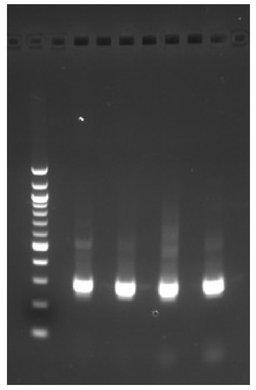
Deliver sgRNAs using AAV to aging-vulnerable brain regions in newborn mice. Connect perturbation identity with single-cell genomic measurements.

Small-Molecule Interventions

Screen epigenetic regulators and approved drugs, then assess 3D genome and transcriptomic responses.

Guide-Region PCR Development

The guide-containing DNA region remains amplifiable across fixation, digestion, and ligation in this preliminary assay.

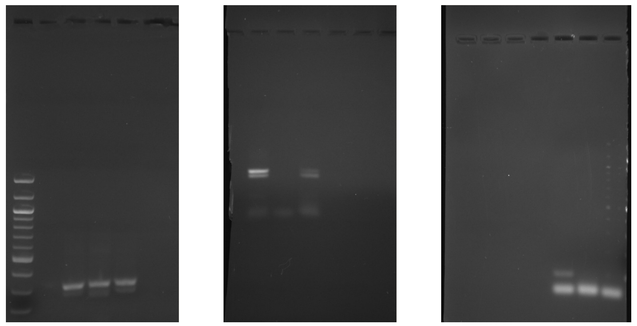


Sample-preparation stages

Lanes from left to right:
100 bp ladder
After ligation
After digestion
After fixation
Fresh sample

Visible products across the four conditions support continued development of guide identification alongside Dip-C sample preparation.

Serial Dilution of Ligated DNA



Dilution inputs: 12.5 ng, 3.1 ng, 780 pg, 195 pg, 50 pg, 12 pg, and 3 pg. These gels test amplification as DNA input decreases.

Toward a Single-Cell Readout

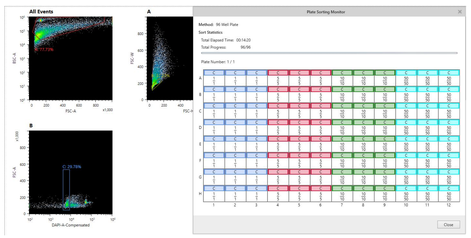


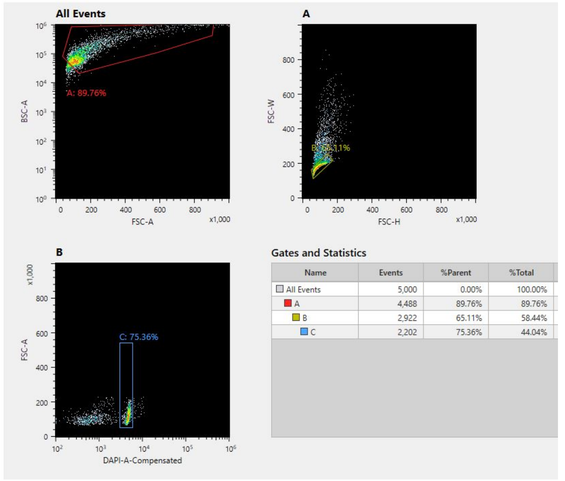
Plate-sorting record and associated PCR gel.



Nuclei Isolation from FFPE Tissue

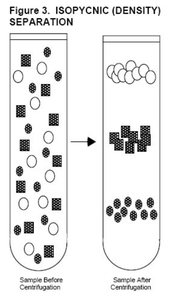
- Formalin-fixed, paraffin-embedded (FFPE) tissue offers a widely used, accessible source of archived biopsy material.
- Efficient separation of nuclei from tissue debris is a sample-preparation challenge for extending Dip-C to FFPE material.

Fluorescence-activated sorting



Example gating sequence from the nuclei-isolation work. Sorting supports nuclei selection and single-nucleus collection, but requires specialized equipment and expertise.

Density-Gradient Enrichment



- Evaluate density-gradient centrifugation as a simpler approach to enriching nuclei.
- Use a layered medium such as iodixanol to separate sample components.
- Optimize gradient composition and centrifugation, then assess nuclei recovery and compatibility with Dip-C.

Next Steps

Refine PCR and nuclei preparation, develop plate-based Dip-C, and establish sequencing and analysis workflows.

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